

EXERCISE 6.9

Q.1 By a rotation of axes, eliminate the xy -term in each of the following equations. Identify the conic and find its elements.

(i) $4x^2 - 4xy + y^2 - 6 = 0$

Solution:

$$4x^2 - 4xy + y^2 - 6 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0$$

we get

$$a = 4, b = 1, 2h = -4$$

As we know that angle of rotation θ is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-4}{4-1}$$

$$\tan 2\theta = -\frac{4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = -\frac{4}{3}$$

$$6 \tan \theta = -4 + 4 \tan^2 \theta$$

$$4 \tan^2 \theta - 6 \tan \theta - 4 = 0$$

$$2 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta - 4 \tan \theta + \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta - 2) + 1 (\tan \theta - 2) = 0$$

$$(2 \tan \theta + 1)(\tan \theta - 2) = 0$$

$$2 \tan \theta + 1 = 0, \tan \theta - 2 = 0$$

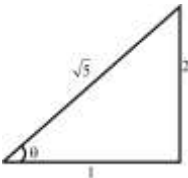
$$\tan \theta = -\frac{1}{2} \text{ and } \tan \theta = 2$$

Neglect $\tan \theta = -\frac{1}{2}$ because θ lies

in first quadrant and in first quadrant $\tan \theta$ is positive.

Now consider $\tan \theta = 2$

From right triangle



$$\sin \theta = \frac{2}{\sqrt{5}}, \cos \theta = \frac{1}{\sqrt{5}}$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

Using values of $\sin \theta$ and $\cos \theta$ we get,

$$x = \frac{X}{\sqrt{5}} - \frac{2Y}{\sqrt{5}} = \frac{X - 2Y}{\sqrt{5}} \dots (ii)$$

$$\text{And } y = X \sin \theta + Y \cos \theta$$

$$y = \frac{2X}{\sqrt{5}} + \frac{Y}{\sqrt{5}} = \frac{2X + Y}{\sqrt{5}} \dots (iv)$$

Put values of x and y in (i), we get

$$\begin{aligned} 4 \left(\frac{X - 2Y}{\sqrt{5}} \right)^2 - 4 \left(\frac{X - 2Y}{\sqrt{5}} \right) \left(\frac{2X + Y}{\sqrt{5}} \right) \\ + \left(\frac{2X + Y}{\sqrt{5}} \right)^2 - 6 = 0 \\ \frac{4(X^2 - 4XY + 4Y^2)}{5} - \frac{4(2X^2 - 3XY - 2Y^2)}{5} \\ + \frac{(4X^2 + 4XY + Y^2)}{5} - 6 = 0 \end{aligned}$$

Multiply by 5

$$\begin{aligned} 4(X^2 - 4XY + 4Y^2) - 4(2X^2 - 3XY - 2Y^2) \\ + (4X^2 + 4XY + Y^2) - 30 = 0 \end{aligned}$$

$$\begin{aligned} 4X^2 - 16XY + 16Y^2 - 8X^2 + 12XY \\ + 8Y^2 + 4X^2 + 4XY + Y^2 - 30 = 0 \end{aligned}$$

$$25Y^2 - 30 = 0 \Rightarrow 25Y^2 = 30$$

$$Y^2 = \frac{30}{25} \Rightarrow Y^2 = \frac{6}{5}$$

So

$$Y = \pm \sqrt{\frac{6}{5}} \text{ are the lines in}$$

XY -plane.

To find their equations in xy -plane, we find Y in terms of ' x ' and ' y '.

Using

$$Y = -x \sin \theta + y \cos \theta$$

Put values of $\sin \theta$ and $\cos \theta$

$$Y = -\frac{2x}{\sqrt{5}} + \frac{y}{\sqrt{5}}$$

$$\text{Put } Y = \pm \sqrt{\frac{6}{5}}$$

$$\pm \frac{\sqrt{6}}{\sqrt{5}} = \frac{-2x+y}{\sqrt{5}}$$

$$\Rightarrow -2x+y = \pm\sqrt{6}$$

$$-2x+y \pm \sqrt{6} = 0$$

$$2x-y \mp \sqrt{6} = 0$$

Hence (i) represents a pair of lines

which are $2x-y+\sqrt{6}=0$ and

$$2x-y-\sqrt{6}=0$$

(ii) $x^2 - 2xy + y^2 - 8x - 8y = 0$

Solution:

$$x^2 - 2xy + y^2 - 8x - 8y = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a=1, b=1, 2h=-2$$

As we know that angle of rotation is

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-2}{1-1} = \frac{-2}{0} = \infty$$

$$\Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

Using equations of transformations

$$x = X \cos \theta - Y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$x = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}} = \frac{X-Y}{\sqrt{2}} \dots (ii)$$

$$\text{And } y = X \sin \theta + Y \cos \theta$$

$$y = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}} = \frac{X+Y}{\sqrt{2}} \dots (iii)$$

Put values of 'x' and 'y' in (i) we get ;

$$\left(\frac{X-Y}{\sqrt{2}}\right)^2 - 2\left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) + \left(\frac{X+Y}{\sqrt{2}}\right)^2 - 8\left(\frac{X-Y}{\sqrt{2}}\right) - 8\left(\frac{X+Y}{\sqrt{2}}\right) = 0$$

$$\frac{(X-Y)^2}{2} - 2\frac{(X^2-Y^2)}{2} + \frac{(X+Y)^2}{2} - 8\frac{(X-Y)}{\sqrt{2}} - 8\frac{(X+Y)}{\sqrt{2}} = 0$$

Multiply by 2 on both sides

$$(X-Y)^2 - 2(X^2-Y^2) + (X+Y)^2$$

$$-8\sqrt{2}(X-Y) - 8\sqrt{2}(X+Y) = 0$$

$$(X+Y)^2 + (X-Y)^2 - 2(X^2-Y^2)$$

$$-8\sqrt{2}X + 8\sqrt{2}Y - 8\sqrt{2}X - 8\sqrt{2}Y = 0$$

$$2(X^2+Y^2) - 2(X^2-Y^2) - 16\sqrt{2}X = 0$$

$$4Y^2 - 16\sqrt{2}X = 0$$

$$Y^2 - 4\sqrt{2}X = 0$$

$$Y^2 = 4\sqrt{2}X \dots (iv)$$

As (iv) is parabola, hence (i)

represents a parabola.

Next we convert 'X' and 'Y' in terms of 'x' and 'y'.

Using equations of transformations

$$X = x \cos \theta + y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$X = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{x+y}{\sqrt{2}}$$

$$X = \frac{x+y}{\sqrt{2}} \dots (v)$$

and

$$Y = -x \sin \theta + y \cos \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$Y = \frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

$$Y = -\frac{(x-y)}{\sqrt{2}} \dots (vi)$$

Elements:

$$\text{From (iv) } Y^2 = 4\sqrt{2}X$$

$$\text{Here } 4a = 4\sqrt{2}$$

$$a = \sqrt{2}$$

$$\text{Focus of (iv) is } F(a, 0) = F(\sqrt{2}, 0)$$

$$X = \sqrt{2} \text{ and } Y = 0$$

$$\text{From (ii) } x = \frac{X-Y}{\sqrt{2}} = \frac{\sqrt{2}-0}{\sqrt{2}} = 1$$

$$\text{From (iii) } y = \frac{X+Y}{\sqrt{2}} = \frac{\sqrt{2}+0}{\sqrt{2}} = 1$$

Hence focus of (i) is (1,1)

Vertex of (iv) is (0,0),

$$X = 0 \text{ and } Y = 0$$

$$\text{From (ii) } x = \frac{X-Y}{\sqrt{2}} = \frac{0-0}{\sqrt{2}} = 0$$

$$\text{From (iii) } y = \frac{X+Y}{\sqrt{2}} = \frac{0+0}{\sqrt{2}} = 0$$

Hence vertex of (i) is (0,0)

Directrix of (iv) is given by

$$X = -a \Rightarrow X = -\sqrt{2}$$

$$X = \frac{x+y}{\sqrt{2}}$$

$$X = -\sqrt{2} \text{ we get}$$

$$-\sqrt{2} = \frac{x+y}{\sqrt{2}} \Rightarrow x+y = -2$$

Hence equation of directrix for (i) is $x+y+2=0$

Axis of (iv) is $Y = 0$

$$\Rightarrow \frac{-x+y}{\sqrt{2}} = 0 \text{ As } Y = \frac{-x+y}{\sqrt{2}}$$

$$-x+y=0$$

$$x-y=0$$

Hence axis of (i) is $x-y=0$

$$\text{(iii) } x^2 + 2xy + y^2 + 2\sqrt{2}x - 2\sqrt{2}y + 2 = 0$$

Solution:

$$x^2 + 2xy + y^2 + 2\sqrt{2}x - 2\sqrt{2}y + 2 = 0 \quad \dots(i)$$

Comparing (i) with

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0,$$

we get

$$a=1, b=1, 2h=2$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{2}{1-1} = \frac{2}{0} = \infty$$

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

As we know that equations of transformation are

$$x = X \cos \theta - Y \sin \theta$$

$$\text{and } y = X \sin \theta + Y \cos \theta$$

For $\theta = 45^\circ$ we have

$$\cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

Thus

$$\begin{cases} x = X \left(\frac{1}{\sqrt{2}} \right) - Y \left(\frac{1}{\sqrt{2}} \right) \\ \text{and} \\ y = X \left(\frac{1}{\sqrt{2}} \right) + Y \left(\frac{1}{\sqrt{2}} \right) \end{cases} \Rightarrow \begin{cases} x = \frac{X-Y}{\sqrt{2}} \dots(ii) \\ \text{and} \\ y = \frac{X+Y}{\sqrt{2}} \dots(iii) \end{cases}$$

Putting (ii) and (iii) in (i), we get

$$\left(\frac{X-Y}{\sqrt{2}} \right)^2 + 2 \left(\frac{X-Y}{\sqrt{2}} \right) \left(\frac{X+Y}{\sqrt{2}} \right) + \left(\frac{X+Y}{\sqrt{2}} \right)^2 + 2\sqrt{2} \left(\frac{X-Y}{\sqrt{2}} \right) - 2\sqrt{2} \left(\frac{X+Y}{\sqrt{2}} \right) + 2 = 0$$

$$\frac{(X-Y)^2}{2} + (X^2 - Y^2) + \frac{(X+Y)^2}{2} + 2(X-Y) - 2(X+Y) + 2 = 0$$

Multiply by 2

$$(X+Y)^2 + (X-Y)^2 + 2(X^2 - Y^2) + 4(X-Y) - 4(X+Y) + 4 = 0$$

$$\Rightarrow 2(X^2 + Y^2) + 2(X^2 - Y^2) + 4X - 4Y - 4X - 4Y + 4 = 0$$

$$\Rightarrow 4X^2 - 8Y + 4 = 0$$

$$\Rightarrow X^2 - 2Y + 1 = 0$$

$$X^2 = 2Y - 1$$

$$X^2 = 2 \left(Y - \frac{1}{2} \right) \dots(iv)$$

Next we find 'X' and 'Y' in terms of 'x' and 'y'

Using equations of transformation

$$X = x \cos \theta + y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$X = x \left(\frac{1}{\sqrt{2}} \right) + y \left(\frac{1}{\sqrt{2}} \right)$$

$$X = \frac{x+y}{\sqrt{2}} \dots (v)$$

and

$$Y = -x \sin \theta + y \cos \theta$$

$$Y = -x \left(\frac{1}{\sqrt{2}} \right) + y \left(\frac{1}{\sqrt{2}} \right)$$

$$Y = \frac{-(x-y)}{\sqrt{2}} \dots (vi)$$

From (iv)

$$X^2 = 2 \left(Y - \frac{1}{2} \right)$$

$$4a = 2 \Rightarrow a = \frac{1}{2}$$

For focus of (iv) we can write

$$F \left(X, Y - \frac{1}{2} \right) = F(0, a)$$

$$X = 0, \quad Y - \frac{1}{2} = a$$

$$X = 0, \quad Y = \frac{1}{2} + \frac{1}{2}$$

$$X = 0, \quad Y = 1$$

So focus of (iv) is (0,1)

$$\text{From (ii)} \quad x = \frac{X-Y}{\sqrt{2}} = \frac{0-1}{\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

$$\text{From (iii)} \quad y = \frac{X+Y}{\sqrt{2}} = \frac{0+1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\text{Hence focus of (i) is } \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

vertex of (iv) is

$$A \left(X, Y - \frac{1}{2} \right) = A(0, 0)$$

$$X = 0, \quad Y - \frac{1}{2} = 0$$

$$X = 0, \quad Y = \frac{1}{2}$$

$$\text{So vertex of (iv) is } \left(0, \frac{1}{2} \right)$$

$$\text{From (ii)} \quad x = \frac{X-Y}{\sqrt{2}} = \frac{0-\frac{1}{2}}{\sqrt{2}} = \frac{-1}{2\sqrt{2}}$$

$$\text{From (iii)} \quad y = \frac{X+Y}{\sqrt{2}} = \frac{0+\frac{1}{2}}{\sqrt{2}} = \frac{1}{2\sqrt{2}}$$

$$\text{Hence vertex of (i) is } \left(\frac{-1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right).$$

Equation of directrix for (iv) is

$$Y - \frac{1}{2} = -a$$

$$Y = -a + \frac{1}{2} \Rightarrow Y = \frac{-1}{2} + \frac{1}{2} = 0$$

Put $Y = 0$ in (vi) we get

$$\frac{-(x-y)}{\sqrt{2}} = 0 \Rightarrow y = x \text{ is the equation}$$

of directrix for (i)

Axis of (vi) is $X = 0$

$$\frac{x+y}{\sqrt{2}} = 0 \quad \text{From (v)}$$

$$x+y=0$$

Hence axis of (i) is $x+y=0$

$$(iv) \quad x^2 + xy + y^2 - 4 = 0$$

Solution:

$$x^2 + xy + y^2 - 4 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a=1, b=1, 2h=1$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b}.$$

$$\tan 2\theta = \frac{1}{0-0} = \frac{1}{0} = \infty$$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

Using equations of transformation

$$x = X \cos \theta - Y \sin \theta$$

$$= X \cos 45^\circ - Y \sin 45^\circ$$

$$= \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}}$$

$$\Rightarrow x = \frac{X - Y}{\sqrt{2}} \dots (ii)$$

And

$$y = X \sin \theta + Y \cos \theta$$

$$= X \sin 45^\circ + Y \cos 45^\circ$$

$$\Rightarrow y = \frac{X + Y}{\sqrt{2}} \dots (iii)$$

Putting values of “x” and “y” in given equation we get

$$\left(\frac{X - Y}{\sqrt{2}}\right)^2 + \left(\frac{X - Y}{\sqrt{2}}\right)\left(\frac{X + Y}{\sqrt{2}}\right) + \left(\frac{X + Y}{\sqrt{2}}\right)^2 - 4 = 0$$

$$\frac{(X - Y)^2}{2} + \frac{(X^2 - Y^2)}{2} + \frac{(X + Y)^2}{2} - 4 = 0$$

$$(X - Y)^2 + (X^2 - Y^2) + (X + Y)^2 - 8 = 0$$

$$[(X + Y)^2 + (X - Y)^2] + (X^2 - Y^2) - 8 = 0$$

$$2(X^2 + Y^2) + X^2 - Y^2 - 8 = 0$$

$$2X^2 + 2Y^2 + X^2 - Y^2 - 8 = 0$$

$$3X^2 + Y^2 - 8 = 0$$

$$3X^2 + Y^2 = 8$$

$$\frac{3X^2}{8} + \frac{Y^2}{8} = 1$$

$$\frac{X^2}{\left(\frac{8}{3}\right)} + \frac{Y^2}{(8)} = 1 \dots (iv)$$

As (iv) represents ellipse, hence (i) is ellipse

Elements:

Here $a^2 = 8, b^2 = \frac{8}{3}$ and

$$c^2 = a^2 - b^2 = 8 - \frac{8}{3} = \frac{16}{3}$$

$$a = 2\sqrt{2}, b = \frac{2\sqrt{2}}{\sqrt{3}}, c = \frac{4}{\sqrt{3}}$$

Next we convert “X” and “Y” in terms of “x” and “y”, so using equations of transformation

$$X = x \cos \theta + y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$X = x\left(\frac{1}{\sqrt{2}}\right) + y\left(\frac{1}{\sqrt{2}}\right)$$

$$X = \frac{x + y}{\sqrt{2}} \dots (v)$$

And

$$Y = -x \sin \theta + y \cos \theta$$

$$Y = -x\left(\frac{1}{\sqrt{2}}\right) + y\left(\frac{1}{\sqrt{2}}\right)$$

$$Y = \frac{-(x - y)}{\sqrt{2}} \dots (vi)$$

Center of (iv) is (0,0)

So coordinates of center are

$$X = 0, Y = 0$$

$$\text{From (ii) } x = \frac{X - Y}{\sqrt{2}} = \frac{0 - 0}{\sqrt{2}} = 0$$

$$\text{From (iii) } y = \frac{X + Y}{\sqrt{2}} = \frac{0 + 0}{\sqrt{2}} = 0$$

Hence center of (i) is (0,0).

Vertices of (iv) are

$$(0, \pm a) = (0, \pm 2\sqrt{2})$$

Here coordinates of vertices are

$$X = 0 \text{ and } Y = \pm 2\sqrt{2}$$

$$\text{For } Y = 2\sqrt{2}, X = 0$$

$$\text{From (ii) } x = \frac{X - Y}{\sqrt{2}} = \frac{0 - 2\sqrt{2}}{\sqrt{2}} = -2$$

$$\text{From (iii) } y = \frac{X+Y}{\sqrt{2}} = \frac{0+2\sqrt{2}}{\sqrt{2}} = 2$$

$$\text{For } Y = -2\sqrt{2}, X = 0$$

$$\text{From (ii) } x = \frac{X-Y}{\sqrt{2}} = \frac{0+2\sqrt{2}}{\sqrt{2}} = 2$$

$$\text{From (iii) } y = \frac{X+Y}{\sqrt{2}} = \frac{0-2\sqrt{2}}{\sqrt{2}} = -2$$

Hence vertices of (i) are $(-2, 2)$ and $(2, -2)$

$$\text{Foci of (iv) are } (0, \pm c) = \left(0, \pm \frac{4}{\sqrt{3}}\right)$$

Here coordinates of foci are

$$X = 0 \text{ and } Y = \pm \frac{4}{\sqrt{3}}$$

$$\text{For } Y = \frac{4}{\sqrt{3}}, X = 0$$

From (ii)

$$x = \frac{X-Y}{\sqrt{2}} = \frac{0 - \frac{4}{\sqrt{3}}}{\sqrt{2}} = \frac{-4}{\sqrt{2}\sqrt{3}} = \frac{-2\sqrt{2}}{\sqrt{3}}$$

From (iii)

$$y = \frac{X+Y}{\sqrt{2}} = \frac{0 + \frac{4}{\sqrt{3}}}{\sqrt{2}} = \frac{4}{\sqrt{2}\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{3}}$$

$$\text{For } Y = \frac{-4}{\sqrt{3}}, X = 0$$

From (ii)

$$x = \frac{X-Y}{\sqrt{2}} = \frac{0 + \frac{4}{\sqrt{3}}}{\sqrt{2}} = \frac{4}{\sqrt{2}\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{3}}$$

From (iii)

$$y = \frac{X+Y}{\sqrt{2}} = \frac{0 - \frac{4}{\sqrt{3}}}{\sqrt{2}} = \frac{-4}{\sqrt{2}\sqrt{3}} = \frac{-2\sqrt{2}}{\sqrt{3}}$$

Here coordinates of foci of (i) are

$$\left(\frac{-2\sqrt{2}}{\sqrt{3}}, \frac{2\sqrt{2}}{\sqrt{3}}\right) \text{ and } \left(\frac{2\sqrt{2}}{\sqrt{3}}, \frac{-2\sqrt{2}}{\sqrt{3}}\right)$$

Major axis of (iv) is $X = 0$

$$\text{From (v) } X = \frac{x+y}{\sqrt{2}}$$

$$0 = \frac{x+y}{\sqrt{2}}$$

$$x+y=0$$

So major axis of (i) is given by the line $y+x=0 \Rightarrow y=-x$

Minor axis of (iv) is $Y = 0$

$$\text{From (vi) } Y = \frac{-(x-y)}{\sqrt{2}}, \text{ so put}$$

$$Y = 0$$

$$\text{We get } \frac{-(x-y)}{\sqrt{2}} = 0 \Rightarrow y = x \text{ which}$$

is the minor axis of (i)

$$\text{Eccentricity of (i) is } e = \frac{c}{a}$$

$$e = \frac{\frac{4}{\sqrt{3}}}{2\sqrt{2}} = \frac{4}{2\sqrt{6}} = \frac{2}{\sqrt{6}}$$

$$(v) \quad 7x^2 - 6\sqrt{3}xy + 13y^2 - 16 = 0$$

Solution:

$$7x^2 - 6\sqrt{3}xy + 13y^2 - 16 = 0 \dots (i)$$

Comparing (i) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 7, b = 13, 2h = -6\sqrt{3}$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-6\sqrt{3}}{7-13} = \frac{-6\sqrt{3}}{-6}$$

$$\tan 2\theta = \sqrt{3} \Rightarrow 2\theta = 60^\circ \Rightarrow \theta = 30^\circ$$

Using equations of transformation

$$x = X \cos \theta - Y \sin \theta, \text{ put } \theta = 30^\circ$$

$$x = X \cos 30^\circ - Y \sin 30^\circ$$

$$x = \frac{X\sqrt{3}}{2} - \frac{Y}{2}$$

$$x = \frac{\sqrt{3}X - Y}{2} \dots \text{(ii)}$$

And

$$y = X \sin \theta + Y \cos \theta, \text{ put } \theta = 30^\circ$$

$$y = X \sin 30^\circ + Y \cos 30^\circ$$

$$y = \frac{X}{2} + \frac{\sqrt{3}Y}{2}$$

$$y = \frac{X + \sqrt{3}Y}{2} \dots \text{(iii)}$$

Putting (ii) and (iii) in (i), we get

$$7\left(\frac{\sqrt{3}X - Y}{2}\right)^2 - 6\sqrt{3}\left(\frac{\sqrt{3}X - Y}{2}\right)\left(\frac{X + \sqrt{3}Y}{2}\right) + 13\left(\frac{X + \sqrt{3}Y}{2}\right)^2 - 16 = 0$$

$$7\left[\frac{3X^2 + Y^2 - 2\sqrt{3}XY}{4}\right] - 6\sqrt{3}\frac{(\sqrt{3}X^2 + 2XY - \sqrt{3}Y^2)}{4} + \frac{13(X^2 + 3Y^2 + 2\sqrt{3}XY)}{4} - 16 = 0$$

$$7(3X^2 + Y^2 - 2\sqrt{3}XY) - 6\sqrt{3}(\sqrt{3}X^2 + 2XY - \sqrt{3}Y^2) + 13(X^2 + 3Y^2 + 2\sqrt{3}XY) - 64 = 0$$

$$21X^2 + 7Y^2 - 14\sqrt{3}XY - 18X^2 - 12\sqrt{3}XY + 18Y^2 + 13X^2 + 39Y^2 + 26\sqrt{3}XY - 64 = 0$$

$$16X^2 + 64Y^2 - 64 = 0$$

$$16X^2 + 64Y^2 = 64$$

$$\frac{X^2}{4} + \frac{Y^2}{1} = 1 \dots \text{(iv)}$$

$$\text{Here } a^2 = 4, b^2 = 1 \text{ and } c^2 = a^2 - b^2 = 4 - 1 = 3$$

$$\Rightarrow a = 2, b = 1, c = \sqrt{3}$$

Next we convert “X” and “Y” in terms of “x” and “y”

Using equations of transformations

$$\begin{cases} X = x \cos \theta + y \sin \theta \\ \text{and} \\ Y = -x \sin \theta + y \cos \theta \end{cases}$$

$$\Rightarrow \begin{cases} X = x \cos 30^\circ + y \sin 30^\circ = \frac{\sqrt{3}x + y}{2} \dots \text{(v)} \\ \text{and} \\ Y = -x \sin 30^\circ + y \cos 30^\circ = \frac{-x + \sqrt{3}y}{2} \dots \text{(vi)} \end{cases}$$

Centre of (iv) is (0,0)

So coordinates of center are $X = 0$
and $Y = 0$

From (ii)

$$x = \frac{\sqrt{3}X - Y}{2} = \frac{\sqrt{3}(0) - 0}{2} = 0$$

From (iii)

$$y = \frac{X + \sqrt{3}Y}{2} = \frac{0 + \sqrt{3}(0)}{2} = 0$$

Hence center of (i) is (0,0).

Vertices of (iv) are $(\pm a, 0) = (\pm 2, 0)$

So coordinates of vertices are

$$X = \pm 2, Y = 0$$

For $X = 2, Y = 0$

From (ii)

$$x = \frac{\sqrt{3}X - Y}{2} = \frac{\sqrt{3}(2) - 0}{2} = \sqrt{3}$$

From (iii)

$$y = \frac{X + \sqrt{3}Y}{2} = \frac{2 + \sqrt{3}(0)}{2} = 1$$

For $X = -2, Y = 0$

From (ii)

$$x = \frac{\sqrt{3}X - Y}{2} = \frac{\sqrt{3}(-2) - 0}{2} = -\sqrt{3}$$

From (iii)

$$y = \frac{X + \sqrt{3}Y}{2} = \frac{-2 + \sqrt{3}(0)}{2} = -1$$

Hence vertices of (i) are $(\sqrt{3}, 1)$ and

Foci of (iv) are $(\pm c, 0) = (\pm \sqrt{3}, 0)$

Here coordinates of foci are

$$X = \pm \sqrt{3}, Y = 0$$

For $X = \sqrt{3}, Y = 0$

From (ii)

$$x = \frac{\sqrt{3}X - Y}{2} = \frac{\sqrt{3}(\sqrt{3}) - 0}{2} = \frac{3}{2}$$

From (iii)

$$y = \frac{X + \sqrt{3}Y}{2} = \frac{\sqrt{3} + \sqrt{3}(0)}{2} = \frac{\sqrt{3}}{2}$$

For $X = -\sqrt{3}, Y = 0$

From (ii)

$$x = \frac{\sqrt{3}X - Y}{2} = \frac{\sqrt{3}(-\sqrt{3}) - 0}{2} = -\frac{3}{2}$$

From (iii)

$$y = \frac{X + \sqrt{3}Y}{2} = \frac{-\sqrt{3} + \sqrt{3}(0)}{2} = -\frac{\sqrt{3}}{2}$$

Hence foci of (i) are $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$ and

$$\left(-\frac{3}{2}, -\frac{\sqrt{3}}{2}\right)$$

Minor axis of (iv) is $X = 0$

Put value of X in (v), we get

$$\frac{\sqrt{3}x + y}{2} = 0 \Rightarrow y = -\sqrt{3}x \text{ is the}$$

Equation of minor axis of (i)

Major axis of (iv) is $Y = 0$

Put value of Y in (vi), we get

$$\frac{-x + \sqrt{3}y}{2} = 0 \Rightarrow -x + \sqrt{3}y = 0$$

$$y = \frac{1}{\sqrt{3}}x \text{ is the equation of major}$$

axis of (i)

Eccentricity of (i) is $e = \frac{c}{a}$

$$\Rightarrow e = \frac{\sqrt{3}}{2}$$

$$\text{(vi)} \quad 4x^2 - 4xy + 7y^2 + 12x + 6y - 9 = 0$$

Solution:

$$4x^2 - 4xy + 7y^2 + 12x + 6y - 9 = 0$$

(1)

Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + C = 0,$$

we get

$$a = 4, b = 7, 2h = -4$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b}$$

$$\tan 2\theta = \frac{-4}{4-7} = \frac{-4}{-3} = \frac{4}{3}$$

$$\tan 2\theta = \frac{4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{3}$$

$$6 \tan \theta = 4 - 4 \tan^2 \theta$$

$$4 \tan^2 \theta + 6 \tan \theta - 4 = 0$$

$$2 \tan^2 \theta + 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta + 4 \tan \theta - \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta + 2) - 1(\tan \theta + 2) = 0$$

$$(2 \tan \theta - 1)(\tan \theta + 2) = 0$$

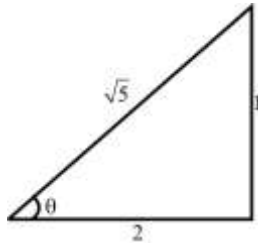
$$2 \tan \theta - 1 = 0, \tan \theta + 2 = 0$$

$$\tan \theta = \frac{1}{2}, \tan \theta = -2$$

As θ lies in 1st quadrant and in 1st quadrant $\tan \theta$ is positive, so neglect $\tan \theta = -2$

$$\text{Now consider } \tan \theta = \frac{1}{2}$$

From right triangle



$$\sin \theta = \frac{1}{\sqrt{5}}, \cos \theta = \frac{2}{\sqrt{5}}$$

Using equations of transformation

$$x = X \cos \theta - Y \sin \theta$$

Using values of $\sin \theta$ and $\cos \theta$ we get;

$$x = \frac{2X}{\sqrt{5}} - \frac{Y}{\sqrt{5}}$$

$$x = \frac{2X - Y}{\sqrt{5}} \quad (2)$$

$$\text{And } y = X \sin \theta + Y \cos \theta$$

Using values of $\sin \theta$ and $\cos \theta$ we get;

$$y = \frac{X}{\sqrt{5}} + \frac{2Y}{\sqrt{5}}$$

$$y = \frac{X + 2Y}{\sqrt{5}} \quad (3)$$

Put values of x and y in (1)

$$4x^2 - 4xy + 7y^2 + 12x + 6y - 9 = 0$$

$$4 \left(\frac{2X - Y}{\sqrt{5}} \right)^2 - 4 \left(\frac{2X - Y}{\sqrt{5}} \right)$$

$$+ 7 \left(\frac{X + 2Y}{\sqrt{5}} \right)^2 + 12 \left(\frac{2X - Y}{\sqrt{5}} \right)$$

$$+ 6 \left(\frac{X + 2Y}{\sqrt{5}} \right) - 9 = 0$$

$$\frac{4(4X^2 + Y^2 - 4XY)}{5} - 4 \frac{(2X^2 + 3XY - 2Y^2)}{5} + 7 \frac{(X^2 + 4Y^2 + 4XY)}{5}$$

$$+ \frac{12(2X - Y)}{\sqrt{5}} + \frac{6(X + 2Y)}{\sqrt{5}} - 9 = 0$$

Multiplying by (5) we get

$$4(4X^2 + Y^2 - 4XY) - 4(2X^2 + 3XY - 2Y^2) + 7(X^2 + 4Y^2 + 4XY)$$

$$+ 12\sqrt{5}(2X - Y) + 6\sqrt{5}(X + 2Y) - 45 = 0$$

$$(16 - 8 + 7)X^2 + (4 + 8 + 28)Y^2 + (-16 - 12 + 28)XY$$

$$+ (24\sqrt{5} + 6\sqrt{5})X + (-12\sqrt{5} + 12\sqrt{5})Y - 45 = 0$$

$$\Rightarrow 15X^2 + 40Y^2 + 30\sqrt{5}X - 45 = 0$$

$$\Rightarrow 3X^2 + 8Y^2 + 6\sqrt{5}X - 9 = 0$$

$$\Rightarrow 3(X^2 + 2\sqrt{5}X) + 8Y^2 - 9 = 0$$

$$3 \left(X^2 + 2\sqrt{5}X + (\sqrt{5})^2 - (\sqrt{5})^2 \right) + 8Y^2 - 9 = 0$$

$$3 \left[(X + \sqrt{5})^2 - 5 \right] + 8Y^2 - 9 = 0$$

$$3(X + \sqrt{5})^2 - 15 + 8Y^2 - 9 = 0$$

$$3(X + \sqrt{5})^2 + 8Y^2 = 24$$

$$\frac{(X + \sqrt{5})^2}{8} + \frac{Y^2}{3} = 1 \quad (4)$$

Here

$$a^2 = 8, b^2 = 3, c^2 = a^2 - b^2 = 8 - 3 = 5$$

$$a = 2\sqrt{2}, b = 3, c = \sqrt{5}$$

As (4) represents ellipse, hence (1) is ellipse

Next we convert X and Y in terms of x and y . So using equations of transformation

$$X = x \cos \theta + y \sin \theta$$

Using values of $\sin \theta$ and $\cos \theta$, we get

$$X = \frac{2x}{\sqrt{5}} + \frac{y}{\sqrt{5}}$$

$$X = \frac{2x+y}{\sqrt{5}} \quad (5)$$

$$Y = -x \sin \theta + y \cos \theta$$

Using values of $\sin \theta$ and $\cos \theta$, we get

$$Y = \frac{-x}{\sqrt{5}} + \frac{2y}{\sqrt{5}}$$

$$Y = \frac{-x+2y}{\sqrt{5}} \quad (6)$$

Elements

Center:

For center of (4), we can write

$$(X + \sqrt{5}, Y) = (0, 0)$$

$$X + \sqrt{5} = 0, \quad Y = 0$$

$$X = -\sqrt{5}, \quad Y = 0$$

Center of (4) is $C(-\sqrt{5}, 0)$

From (2)

$$x = \frac{2X - Y}{\sqrt{5}} = \frac{2(-\sqrt{5}) - 0}{\sqrt{5}} = -2$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{-\sqrt{5} + 2(0)}{\sqrt{5}} = -1$$

Hence center of (1) is $(-2, -1)$

Major Axis:

Equation of major axis of (4) is

$Y = 0$, put $Y = 0$ in (6) we get

$$\frac{-x+2y}{\sqrt{5}} = 0 \Rightarrow -x+2y = 0$$

$$y = \frac{1}{2}x \text{ is the equation of major axis}$$

of (1)

Minor axis:

Equation of minor axis of (4) is

$$X + \sqrt{5} = 0 \Rightarrow X = -\sqrt{5}$$

Put $X = -\sqrt{5}$ in (5) we get

$$\frac{2x+y}{\sqrt{5}} = -\sqrt{5}$$

$$2x+y = -5$$

$2x+y+5=0$ is the equation of minor axis of (1)

Vertices:

For vertices of (4) we can write

$$(X + \sqrt{5}, Y) = (\pm a, 0)$$

$$X + \sqrt{5} = \pm a, \quad Y = 0$$

$$X + \sqrt{5} = \pm 2\sqrt{2}, \quad Y = 0$$

$$X = -\sqrt{5} \pm 2\sqrt{2}, \quad Y = 0$$

So vertices of (4) are $(-\sqrt{5} \pm 2\sqrt{2}, 0)$

For $X = -\sqrt{5} + 2\sqrt{2}, Y = 0$

From (2)

$$\begin{aligned} x &= \frac{2X - Y}{\sqrt{5}} = \frac{2(-\sqrt{5} + 2\sqrt{2}) - 0}{\sqrt{5}} \\ &= \frac{-2\sqrt{5} + 4\sqrt{2}}{\sqrt{5}} = \frac{-10 + 4\sqrt{10}}{5} \end{aligned}$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{-\sqrt{5} + 2\sqrt{2} - 2(0)}{\sqrt{5}} = \frac{-5 + 2\sqrt{10}}{5}$$

For $X = -\sqrt{5} - 2\sqrt{2}, Y = 0$

From (2)

$$\begin{aligned} x &= \frac{2X - Y}{\sqrt{5}} = \frac{2(-\sqrt{5} - 2\sqrt{2}) - 0}{\sqrt{5}} \\ &= \frac{-2\sqrt{5} - 4\sqrt{2}}{\sqrt{5}} = \frac{-10 - 4\sqrt{10}}{5} \end{aligned}$$

From (3)

$$y = \frac{X+2Y}{\sqrt{5}} = \frac{-\sqrt{5}-2\sqrt{2}-2(0)}{\sqrt{5}} = \frac{-5-2\sqrt{10}}{5}$$

Hence vertices of (1) are

$$\left(\frac{-10+4\sqrt{10}}{5}, \frac{-5+2\sqrt{10}}{5} \right) \text{ and}$$

$$\left(\frac{-10-4\sqrt{10}}{5}, \frac{-5-2\sqrt{10}}{5} \right)$$

$$\text{Or } \left(-2 + \sqrt{\frac{32}{5}}, -1 + \sqrt{\frac{8}{5}} \right) \text{ and}$$

$$\left(-2 - \sqrt{\frac{32}{5}}, -1 - \sqrt{\frac{8}{5}} \right)$$

Foci:

For foci of (4) we can write

$$(X + \sqrt{5}, Y) = (\pm c, 0)$$

$$X + \sqrt{5} = \pm c, \quad Y = 0$$

$$X + \sqrt{5} = \pm \sqrt{5}, \quad Y = 0$$

$$X = -\sqrt{5} \pm \sqrt{5}, \quad Y = 0$$

$$X = 0, -2\sqrt{5}, \quad Y = 0$$

So foci of (4) are (0,0) and

$$(-2\sqrt{5}, 0)$$

For $X = 0, Y = 0$

From (2)

$$x = \frac{2X - Y}{\sqrt{5}} = \frac{2(0) - 0}{\sqrt{5}} = \frac{0}{\sqrt{5}} = 0$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{0 - 2(0)}{\sqrt{5}} = 0$$

For $X = -2\sqrt{5}, Y = 0$

From (2)

$$x = \frac{2X - Y}{\sqrt{5}} = \frac{2(-2\sqrt{5}) - 0}{\sqrt{5}} = -4$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{-2\sqrt{5} - 2(0)}{\sqrt{5}} = -2$$

Hence foci of (1) are (0,0) and (-4,-2)

Eccentricity:

Eccentricity of (1) is

$$e = \frac{c}{a} = \frac{\sqrt{5}}{2\sqrt{2}} = \sqrt{\frac{5}{8}}$$

$$\text{(vii) } xy - 4x - 2y = 0$$

Solution: Consider

$$xy - 4x - 2y = 0 \quad (1)$$

Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 0, b = 0, 2h = 1$$

We know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{1}{0-0} = \frac{1}{0} = \infty$$

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

Using equations of transformation

$$x = X \cos \theta - Y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$x = X \left(\frac{1}{\sqrt{2}} \right) - Y \left(\frac{1}{\sqrt{2}} \right)$$

$$x = \frac{X - Y}{\sqrt{2}} \quad (2)$$

And $y = X \sin \theta + Y \cos \theta$

$$y = X \left(\frac{1}{\sqrt{2}} \right) + Y \left(\frac{1}{\sqrt{2}} \right)$$

$$y = \frac{X + Y}{\sqrt{2}} \quad (3)$$

Putting values of x and y in (1), we get

$$\left(\frac{X - Y}{\sqrt{2}} \right) \left(\frac{X + Y}{\sqrt{2}} \right) - 4 \left(\frac{X - Y}{\sqrt{2}} \right) - 2 \left(\frac{X + Y}{\sqrt{2}} \right) = 0$$

$$\frac{X^2 - Y^2}{2} - \frac{4(X - Y)}{\sqrt{2}} - \frac{2(X + Y)}{\sqrt{2}} = 0$$

Multiply both sides by 2

$$X^2 - Y^2 - 4\sqrt{2}(X - Y) - 2\sqrt{2}(X + Y) = 0$$

$$X^2 - Y^2 - 6\sqrt{2}X + 2\sqrt{2}Y = 0$$

$$\begin{aligned}
 & (X^2 - 6\sqrt{2}X) - (Y^2 - 2\sqrt{2}Y) = 0 \\
 \Rightarrow & (X^2 - 6\sqrt{2}X + (3\sqrt{2})^2 - (3\sqrt{2})^2) \\
 & - (Y^2 - 2\sqrt{2}Y + (\sqrt{2})^2 - (\sqrt{2})^2) = 0 \\
 \Rightarrow & (X - 3\sqrt{2})^2 - 18 - (Y - \sqrt{2})^2 + 2 = 0 \\
 \Rightarrow & (X - 3\sqrt{2})^2 - (Y - \sqrt{2})^2 = 16 \\
 \Rightarrow & \frac{(X - 3\sqrt{2})^2}{16} - \frac{(Y - \sqrt{2})^2}{16} = 1 \dots (4)
 \end{aligned}$$

As (4) represents hyperbola, hence (1) is hyperbola.

Here $a^2 = 16, b^2 = 16$ and

$$c^2 = a^2 + b^2 = 32$$

$$\Rightarrow a = 4, b = 4, c = 4\sqrt{2}$$

Now we convert X and Y in terms of x and y . Using equations of transformation

$$X = x \cos \theta + y \sin \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$X = \frac{x+y}{\sqrt{2}} \quad (5)$$

$$\text{And } Y = -x \sin \theta + y \cos \theta$$

For

$$\theta = 45^\circ \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$Y = \frac{-x+y}{\sqrt{2}} \quad (6)$$

Elements:

Center:

For center of (4), we can write

$$X - 3\sqrt{2} = 0, \quad Y - \sqrt{2} = 0$$

$$X = 3\sqrt{2}, \quad Y = \sqrt{2}$$

So center of (4) is $(3\sqrt{2}, \sqrt{2})$

$$\text{From (2) } x = \frac{X - Y}{\sqrt{2}} = \frac{3\sqrt{2} - \sqrt{2}}{\sqrt{2}} = 2$$

$$\text{From (3) } y = \frac{X + Y}{\sqrt{2}} = \frac{3\sqrt{2} + \sqrt{2}}{\sqrt{2}} = 4$$

Hence center of (1) is (2, 4)

Transverse axis:

Equation of transverse axis of (4) is

$$Y - \sqrt{2} = 0 \Rightarrow Y = \sqrt{2}$$

Put value of Y in (6)

$$\frac{-x+y}{\sqrt{2}} = \sqrt{2} \Rightarrow -x+y=2$$

$x - y + 2 = 0$ is the equation of transverse axis for (1)

Conjugate axis:

Conjugate axis of (4) is given by

$$X - 3\sqrt{2} = 0 \Rightarrow X = 3\sqrt{2}$$

Put value of X in (5)

$$\frac{x+y}{\sqrt{2}} = 3\sqrt{2}$$

$$x+y=6$$

$x + y - 6 = 0$ is the equation of conjugate axis for (1)

Vertices:

For vertices of (4), we can write

$$X - 3\sqrt{2} = \pm a, \quad Y - \sqrt{2} = 0$$

$$X - 3\sqrt{2} = \pm 4, \quad Y - \sqrt{2} = 0$$

$$X = 3\sqrt{2} \pm 4, \quad Y = \sqrt{2}$$

So vertices of (4) are $(3\sqrt{2} \pm 4, \sqrt{2})$

$$\text{For } X = 3\sqrt{2} + 4, Y = \sqrt{2}$$

From (2)

$$x = \frac{X - Y}{\sqrt{2}} = \frac{3\sqrt{2} + 4 - \sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2} + 4}{\sqrt{2}} = 2 + 2\sqrt{2}$$

From (3)

$$y = \frac{X + Y}{\sqrt{2}} = \frac{3\sqrt{2} + 4 + \sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2} + 4}{\sqrt{2}} = 4 + 2\sqrt{2}$$

$$\text{For } X = 3\sqrt{2} - 4, Y = \sqrt{2}$$

From (2)

$$\begin{aligned}
 x &= \frac{X - Y}{\sqrt{2}} = \frac{3\sqrt{2} - 4 - \sqrt{2}}{\sqrt{2}} \\
 &= \frac{2\sqrt{2} - 4}{\sqrt{2}} = 2 - 2\sqrt{2}
 \end{aligned}$$

From (3)

$$y = \frac{X+Y}{\sqrt{2}} = \frac{3\sqrt{2}-4+\sqrt{2}}{\sqrt{2}}$$

$$= \frac{4\sqrt{2}-4}{\sqrt{2}} = 4-2\sqrt{2}$$

Hence vertices of (1) are
 $(2+2\sqrt{2}, 4+2\sqrt{2})$ and
 $(2-2\sqrt{2}, 4-2\sqrt{2})$

Foci:

For foci of (4), we can write

$$X-3\sqrt{2} = \pm c, \quad Y-\sqrt{2} = 0$$

$$X-3\sqrt{2} = \pm 4\sqrt{2}, \quad Y-\sqrt{2} = 0$$

$$X = 3\sqrt{2} \pm 4\sqrt{2}, \quad Y = \sqrt{2}$$

$$X = 7\sqrt{2}, -\sqrt{2}, \quad Y = \sqrt{2}$$

So foci of (4) are $(7\sqrt{2}, \sqrt{2})$ and
 $(-\sqrt{2}, \sqrt{2})$

For $X = 7\sqrt{2}, Y = \sqrt{2}$

From (2)

$$x = \frac{X-Y}{\sqrt{2}} = \frac{7\sqrt{2}-\sqrt{2}}{\sqrt{2}} = \frac{6\sqrt{2}}{\sqrt{2}} = 6$$

From (3)

$$y = \frac{X+Y}{\sqrt{2}} = \frac{7\sqrt{2}+\sqrt{2}}{\sqrt{2}} = \frac{8\sqrt{2}}{\sqrt{2}} = 8$$

For $X = -\sqrt{2}, Y = \sqrt{2}$

From (2)

$$x = \frac{X-Y}{\sqrt{2}} = \frac{-\sqrt{2}-\sqrt{2}}{\sqrt{2}} = \frac{-2\sqrt{2}}{\sqrt{2}} = -2$$

From (3)

$$y = \frac{X+Y}{\sqrt{2}} = \frac{-\sqrt{2}+\sqrt{2}}{\sqrt{2}} = \frac{0}{\sqrt{2}} = 0$$

Hence foci of (1) are (6,8) and
 $(-2,0)$

Eccentricity:

Eccentricity of (1) is

$$e = \frac{c}{a} = \frac{4\sqrt{2}}{4} = \sqrt{2} > 1$$

$$(viii) \quad x^2 + 4xy - 2y^2 - 6 = 0$$

Solution: Consider

$$x^2 + 4xy - 2y^2 - 6 = 0 \quad (1)$$

Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a=1, b=-2, 2h=4$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{4}{1-(-2)} = \frac{4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{3}$$

$$6 \tan \theta = 4 - 4 \tan^2 \theta$$

$$4 \tan^2 \theta + 6 \tan \theta - 4 = 0$$

$$2 \tan^2 \theta + 3 \tan \theta - 2 = 0$$

(Dividing by 2)

$$2 \tan^2 \theta + 4 \tan \theta - \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta + 2) - 1(\tan \theta + 2) = 0$$

$$(2 \tan \theta - 1)(\tan \theta + 2) = 0$$

$$2 \tan \theta - 1 = 0, \tan \theta + 2 = 0$$

$$\tan \theta = \frac{1}{2}, \tan \theta = -2$$

As $0 < \theta < 90^\circ$ and $\tan \theta$ is positive in 1st quadrant, so neglect $\tan \theta = -2$

$$\text{Consider } \tan \theta = \frac{1}{2}$$

From the right triangle

$$\sin \theta = \frac{1}{\sqrt{5}}, \cos \theta = \frac{2}{\sqrt{5}}$$

Using equations of transformation

$$x = Y \cos \theta - Y \sin \theta$$

Using values of $\sin \theta$ and $\cos \theta$, we get

$$x = \frac{2X}{\sqrt{5}} - \frac{Y}{\sqrt{5}} = \frac{2X-Y}{\sqrt{5}} \quad (2)$$

And $y = X \sin \theta + Y \cos \theta$

$$y = \frac{X}{\sqrt{5}} + \frac{2Y}{\sqrt{5}}$$

$$y = \frac{X+2Y}{\sqrt{5}} \quad (3)$$

Substituting values of x and y in (1), we get

$$\left(\frac{2X-Y}{\sqrt{5}}\right)^2 + 4\left(\frac{2X-Y}{\sqrt{5}}\right) - 2\left(\frac{X+2Y}{\sqrt{5}}\right)^2 - 6 = 0$$

$$\left(\frac{4X^2+Y^2-4XY}{5}\right) + 4\left(\frac{2X^2+3XY-2Y^2}{5}\right) - 2\left(\frac{X^2+4Y^2+4XY}{5}\right)^2 - 6 = 0$$

Multiplying by 5, we get

$$4X^2 + Y^2 - 4XY + 8X^2 + 12XY - 8Y^2 - 2X^2 - 8Y^2 - 8XY - 30 = 0$$

$$10X^2 - 15Y^2 - 30 = 0$$

$$2X^2 - 3Y^2 - 6 = 0$$

$$2X^2 - 3Y^2 = 6$$

$$\frac{X^2}{3} - \frac{Y^2}{2} = 1 \quad (4)$$

Which represents a hyperbola, hence (1) represents hyperbola

Here

$$a^2 = 3, b^2 = 2, c^2 = a^2 + b^2 = 2 + 3 = 5$$

$$\Rightarrow a = \sqrt{3}, b = \sqrt{2}, c = \sqrt{5}$$

Next we convert X and Y in terms of x and y

$$\text{As } X = x \cos \theta + y \sin \theta$$

$$X = \frac{2x}{\sqrt{5}} + \frac{y}{\sqrt{5}}$$

$$X = \frac{2x+y}{\sqrt{5}} \quad (5)$$

$$\text{And } Y = -x \sin \theta + y \cos \theta$$

$$Y = \frac{-x}{\sqrt{5}} + \frac{2y}{\sqrt{5}}$$

$$Y = \frac{-x+2y}{\sqrt{5}} \quad (6)$$

Elements:

Center of (4) is (0,0) so coordinates of center of (4) are $X = 0, Y = 0$

From (2)

$$x = \frac{2X-Y}{\sqrt{5}} = \frac{2(0)-(0)}{\sqrt{5}} = 0$$

From (3)

$$y = \frac{X+2Y}{\sqrt{5}} = \frac{0+2(0)}{\sqrt{5}} = 0$$

Hence center of (1) is (0,0)

Transvers axis:

Transvers axis of (4) is $Y = 0$

$$\text{Put } Y = 0 \text{ in (6), we get } \frac{-x+2y}{\sqrt{5}} = 0$$

$$\Rightarrow -x+2y=0 \Rightarrow y = \frac{1}{2}x \text{ is the}$$

transvers axis of (1)

Conjugate axis:

Conjugate axis of (4) is $X = 0$

$$\text{Put } X = 0 \text{ in (5), we get } \frac{2x+y}{\sqrt{5}} = 0$$

$$\Rightarrow 2x+y=0 \Rightarrow y = -2x \text{ is the equation of conjugate axis of (1)}$$

Vertices:

$$\text{Vertices of (4) are } (\pm a, 0) = (\pm \sqrt{3}, 0)$$

Here coordinates of vertices are

$$X = \pm \sqrt{3} \text{ and } Y = 0$$

$$\text{For } X = \sqrt{3}, Y = 0$$

From (2)

$$x = \frac{2X-Y}{\sqrt{5}} = \frac{2(\sqrt{3})-(0)}{\sqrt{5}} = 2\sqrt{\frac{3}{5}}$$

From (3)

$$y = \frac{X+2Y}{\sqrt{5}} = \frac{\sqrt{3}+2(0)}{\sqrt{5}} = \sqrt{\frac{3}{5}}$$

$$\text{For } X = -\sqrt{3}, Y = 0$$

From (2)

$$x = \frac{2X-Y}{\sqrt{5}} = \frac{2(-\sqrt{3})-(0)}{\sqrt{5}} = -2\sqrt{\frac{3}{5}}$$

From (3)

$$y = \frac{X+2Y}{\sqrt{5}} = \frac{-\sqrt{3}+2(0)}{\sqrt{5}} = -\sqrt{\frac{3}{5}}$$

Hence vertices of (1) are

$$\left(2\sqrt{\frac{3}{5}}, \sqrt{\frac{3}{5}}\right) \text{ and } \left(-2\sqrt{\frac{3}{5}}, -\sqrt{\frac{3}{5}}\right)$$

Foci:

Foci of (4) are $(\pm c, 0) = (\pm\sqrt{5}, 0)$

Here coordinates of foci are

$$X = \pm\sqrt{5}, Y = 0$$

For $X = \sqrt{5}, Y = 0$

From (2)

$$x = \frac{2X - Y}{\sqrt{5}} = \frac{2(\sqrt{5}) - (0)}{\sqrt{5}} = 2$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{\sqrt{5} + 2(0)}{\sqrt{5}} = 1$$

For $X = -\sqrt{5}, Y = 0$

From (2)

$$x = \frac{2X - Y}{\sqrt{5}} = \frac{2(-\sqrt{5}) - (0)}{\sqrt{5}} = -2$$

From (3)

$$y = \frac{X + 2Y}{\sqrt{5}} = \frac{-\sqrt{5} + 2(0)}{\sqrt{5}} = -1$$

Hence foci of (1) are $(2, 1), (-2, -1)$

Eccentricity:

$$e = \frac{c}{a} = \frac{\sqrt{5}}{\sqrt{3}} = \sqrt{\frac{5}{3}} \text{ is the eccentricity}$$

of (1)

$$(ix) \quad x^2 - 4xy - 2y^2 + 10x + 4y = 0$$

Solution: Consider

$$x^2 - 4xy - 2y^2 + 10x + 4y = 0 \quad (1)$$

Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 1, b = -2, 2h = -4$$

As we know that angle of rotation is given by

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-4}{1+2} = -\frac{4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{-4}{3}$$

$$\frac{\tan \theta}{1 - \tan^2 \theta} = \frac{-2}{3}$$

$$3 \tan \theta = -2 + 2 \tan^2 \theta$$

$$2 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta - 4 \tan \theta + \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta - 2) + 1(\tan \theta - 2) = 0$$

$$(\tan \theta - 2)(2 \tan \theta + 1) = 0$$

$$\tan \theta - 2 = 0, 2 \tan \theta + 1 = 0$$

$$\tan \theta = 2, \tan \theta = -\frac{1}{2}$$

As $0 < \theta < 90^\circ$ and $\tan \theta$ is positive in 1st quadrant, so neglect

$$\tan \theta = -\frac{1}{2}$$

Consider $\tan \theta = 2$

From the right triangle

$$\sin \theta = \frac{2}{\sqrt{5}}, \cos \theta = \frac{1}{\sqrt{5}}$$

Using equation of transformation

$$x = X \cos \theta - Y \sin \theta$$

$$x = \frac{X}{\sqrt{5}} - \frac{2Y}{\sqrt{5}}$$

$$x = \frac{X - 2Y}{\sqrt{5}} \quad (2)$$

And $y = X \sin \theta + Y \cos \theta$

$$y = \frac{2X}{\sqrt{5}} + \frac{Y}{\sqrt{5}}$$

$$y = \frac{2X + Y}{\sqrt{5}} \quad (3)$$

Put (2) and (3) in (1), we get

$$\left(\frac{X-2Y}{\sqrt{5}}\right)^2 - 4\left(\frac{X-2Y}{\sqrt{5}}\right)\left(\frac{2X+Y}{\sqrt{5}}\right) - 2\left(\frac{2X+Y}{\sqrt{5}}\right)^2 + 10\left(\frac{X-2Y}{\sqrt{5}}\right) + 4\left(\frac{2X+Y}{\sqrt{5}}\right) = 0$$

$$\frac{(X^2 + 4Y^2 - 4XY)}{5} - 4\frac{(2X^2 - 3XY - 2Y^2)}{5} - 2\frac{(4X^2 + Y^2 + 4XY)}{5}$$

$$+ 10\frac{(X-2Y)}{\sqrt{5}} + 4\left(\frac{2X+Y}{\sqrt{5}}\right) = 0$$

Multiplying by 5, we get

$$X^2 + 4Y^2 - 4XY - 8X^2 + 12XY$$

$$+ 8Y^2 - 8X^2 - 2Y^2 - 8XY + 10\sqrt{5}X$$

$$- 20\sqrt{5}Y + 8\sqrt{5}X + 4\sqrt{5}Y = 0$$

$$-15X^2 + 10Y^2 + 18\sqrt{5}X - 16\sqrt{5}Y = 0$$

$$10Y^2 - 16\sqrt{5}Y - 15X^2 + 18\sqrt{5}X = 0$$

$$10\left(Y^2 - \frac{8}{\sqrt{5}}Y\right) - 15\left(X^2 - \frac{6}{\sqrt{5}}X\right) = 0$$

$$10\left(Y^2 - \frac{8}{\sqrt{5}}Y + \left(\frac{4}{\sqrt{5}}\right)^2 - \left(\frac{4}{\sqrt{5}}\right)^2\right)$$

$$-15\left(X^2 - \frac{6}{\sqrt{5}}X + \left(\frac{3}{\sqrt{5}}\right)^2 - \left(\frac{3}{\sqrt{5}}\right)^2\right) = 0$$

$$10\left(\left(Y - \frac{4}{\sqrt{5}}\right)^2 - \frac{16}{5}\right)$$

$$-15\left(\left(X - \frac{3}{\sqrt{5}}\right)^2 - \frac{9}{5}\right) = 0$$

$$10\left(Y - \frac{4}{\sqrt{5}}\right)^2 - 15\left(X - \frac{3}{\sqrt{5}}\right)^2 - 32 + 27 = 0$$

$$10\left(Y - \frac{4}{\sqrt{5}}\right)^2 - 15\left(X - \frac{3}{\sqrt{5}}\right)^2 = 5$$

$$\frac{\left(Y - \frac{4}{\sqrt{5}}\right)^2}{\frac{1}{2}} - \frac{\left(X - \frac{3}{\sqrt{5}}\right)^2}{\frac{1}{3}} = 1 \dots (4)$$

(4) represents hyperbola, hence (1) represents hyperbola

$$\text{Here } a^2 = \frac{1}{2}, b^2 = \frac{1}{3}, c^2 = a^2 + b^2 = \frac{5}{6}$$

$$\Rightarrow a = \frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{3}}, c = \sqrt{\frac{5}{6}}$$

Next we convert X and Y in terms of x and y . Using equations of transformation

$$X = x \cos \theta + y \sin \theta$$

$$X = \frac{x}{\sqrt{5}} + \frac{2y}{\sqrt{5}}$$

$$X = \frac{x+2y}{\sqrt{5}} \quad (5)$$

$$\text{And } Y = -x \sin \theta + y \cos \theta$$

$$Y = \frac{-2x}{\sqrt{5}} + \frac{y}{\sqrt{5}}$$

$$Y = \frac{-2x+y}{\sqrt{5}} \quad (6)$$

Elements:

Center:

For center of (4), we can write

$$X - \frac{3}{\sqrt{5}} = 0, \quad Y - \frac{4}{\sqrt{5}} = 0$$

$$X = \frac{3}{\sqrt{5}}, \quad Y = \frac{4}{\sqrt{5}}$$

$$\text{So center of (4) is } \left(\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}}\right)$$

From (2)

$$x = \frac{X - 2Y}{\sqrt{5}} = \frac{\frac{3}{\sqrt{5}} - 2\left(\frac{4}{\sqrt{5}}\right)}{\sqrt{5}} = \frac{3-8}{5} = -1$$

From (3)

$$y = \frac{2X + Y}{\sqrt{5}} = \frac{2\left(\frac{3}{\sqrt{5}}\right) + \frac{4}{\sqrt{5}}}{\sqrt{5}} = \frac{6+4}{5} = 2$$

Hence center of (1) is $(-1, 2)$

Transverse axis:

Transverse axis of (4) is

$$X - \frac{3}{\sqrt{5}} = 0 \Rightarrow X = \frac{3}{\sqrt{5}}$$

$$\text{Put } X = \frac{3}{\sqrt{5}} \text{ in (5), we get}$$

$$\frac{x+2y}{\sqrt{5}} = \frac{3}{\sqrt{5}}$$

$x+2y=3$ is the transverse axis of (1)

Conjugate axis:

Conjugate axis of (4) is

$$Y - \frac{4}{\sqrt{5}} = 0 \Rightarrow Y = \frac{4}{\sqrt{5}}$$

$$\text{Put } Y = \frac{4}{\sqrt{5}} \text{ in (6), we get}$$

$$\frac{-2x+y}{\sqrt{5}} = \frac{4}{\sqrt{5}}$$

$$-2x+y=4$$

$2x - y + 4 = 0$ is the conjugate axis
for (1)

Foci:

For foci of (4), we can write

$$X - \frac{3}{\sqrt{5}} = 0, \quad Y - \frac{4}{\sqrt{5}} = \pm c$$

$$X - \frac{3}{\sqrt{5}} = 0, \quad Y - \frac{4}{\sqrt{5}} = \pm \sqrt{\frac{5}{6}}$$

$$X = \frac{3}{\sqrt{5}}, \quad Y = \frac{4}{\sqrt{5}} \pm \sqrt{\frac{5}{6}}$$

So foci of (4) are given by

$$\left(\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}} \pm \sqrt{\frac{5}{6}} \right)$$

For $X = \frac{3}{\sqrt{5}}, Y = \frac{4}{\sqrt{5}} + \sqrt{\frac{5}{6}}$

From (2)

$$\begin{aligned} x = \frac{X - 2Y}{\sqrt{5}} &= \frac{\frac{3}{\sqrt{5}} - 2\left(\frac{4}{\sqrt{5}} + \sqrt{\frac{5}{6}}\right)}{\sqrt{5}} \\ &= \frac{1}{\sqrt{5}} \left(\frac{3}{\sqrt{5}} - \frac{8}{\sqrt{5}} - \frac{2\sqrt{5}}{\sqrt{6}} \right) \\ &= \frac{3}{5} - \frac{8}{5} - \frac{2}{\sqrt{6}} = -1 - \frac{2}{\sqrt{6}} \end{aligned}$$

From (3)

$$\begin{aligned} y = \frac{2X + Y}{\sqrt{5}} &= \frac{1}{\sqrt{5}} \left(2\left(\frac{3}{\sqrt{5}}\right) + \frac{4}{\sqrt{5}} + \sqrt{\frac{5}{6}} \right) \\ &= \frac{6}{5} + \frac{4}{5} + \frac{1}{\sqrt{6}} = 2 + \frac{1}{\sqrt{6}} \end{aligned}$$

For $X = \frac{3}{\sqrt{5}}, Y = \frac{4}{\sqrt{5}} - \sqrt{\frac{5}{6}}$

From (2)

$$\begin{aligned} x = \frac{X - 2Y}{\sqrt{5}} &= \frac{\frac{3}{\sqrt{5}} - 2\left(\frac{4}{\sqrt{5}} - \sqrt{\frac{5}{6}}\right)}{\sqrt{5}} \\ &= \frac{1}{\sqrt{5}} \left(\frac{3}{\sqrt{5}} - \frac{8}{\sqrt{5}} + \frac{2\sqrt{5}}{\sqrt{6}} \right) \\ &= \frac{3}{5} - \frac{8}{5} + \frac{2}{\sqrt{6}} = -1 + \frac{2}{\sqrt{6}} \end{aligned}$$

$$\begin{aligned} y = \frac{2X + Y}{\sqrt{5}} &= \frac{1}{\sqrt{5}} \left(2\left(\frac{3}{\sqrt{5}}\right) + \frac{4}{\sqrt{5}} - \sqrt{\frac{5}{6}} \right) \\ &= \frac{6}{5} + \frac{4}{5} - \frac{1}{\sqrt{6}} = 2 - \frac{1}{\sqrt{6}} \end{aligned}$$

Hence foci of (1) are

$$\left(-1 - \frac{2}{\sqrt{6}}, 2 + \frac{1}{\sqrt{6}} \right) \text{ and}$$

$$\left(-1 + \frac{2}{\sqrt{6}}, 2 - \frac{1}{\sqrt{6}} \right)$$

Vertices:

For vertices of (4), we can write

$$X - \frac{3}{\sqrt{5}} = 0, \quad Y - \frac{4}{\sqrt{5}} = \pm a$$

$$X - \frac{3}{\sqrt{5}} = 0, \quad Y = \frac{4}{\sqrt{5}} \pm \frac{1}{\sqrt{2}}$$

So vertices of (4) are

$$\left(\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}} \pm \frac{1}{\sqrt{2}} \right)$$

For $X = \frac{3}{\sqrt{5}}, Y = \frac{4}{\sqrt{5}} + \frac{1}{\sqrt{2}}$

From (2)

$$\begin{aligned} x = \frac{X - 2Y}{\sqrt{5}} &= \frac{\frac{3}{\sqrt{5}} - 2\left(\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{2}}\right)}{\sqrt{5}} \\ &= \frac{1}{\sqrt{5}} \left(\frac{3}{\sqrt{5}} - \frac{8}{\sqrt{5}} - \frac{2}{\sqrt{2}} \right) \\ &= \frac{3}{5} - \frac{8}{5} - \frac{2}{\sqrt{10}} = -1 - \frac{2}{\sqrt{10}} \end{aligned}$$

From (3)

$$y = \frac{2X+Y}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left(2 \left(\frac{3}{\sqrt{5}} \right) + \frac{4}{\sqrt{5}} + \frac{1}{\sqrt{2}} \right)$$

$$= \frac{6}{5} + \frac{4}{5} + \frac{1}{\sqrt{10}} = 2 + \frac{1}{\sqrt{10}}$$

$$\text{For } X = \frac{3}{\sqrt{5}}, Y = \frac{4}{\sqrt{5}} - \frac{1}{\sqrt{2}}$$

From (2)

$$x = \frac{X-2Y}{\sqrt{5}} = \frac{\frac{3}{\sqrt{5}} - 2 \left(\frac{4}{\sqrt{5}} - \frac{1}{\sqrt{2}} \right)}{\sqrt{5}}$$

$$= \frac{1}{\sqrt{5}} \left(\frac{3}{\sqrt{5}} - \frac{8}{\sqrt{5}} + \frac{2}{\sqrt{2}} \right)$$

$$= \frac{3}{5} - \frac{8}{5} + \frac{2}{\sqrt{10}} = -1 + \frac{2}{\sqrt{10}}$$

From (3)

$$y = \frac{2X+Y}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left(2 \left(\frac{3}{\sqrt{5}} \right) + \frac{4}{\sqrt{5}} - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{6}{5} + \frac{4}{5} - \frac{1}{\sqrt{10}} = 2 - \frac{1}{\sqrt{10}}$$

Hence vertices of (1) are

$$\left(-1 - \frac{2}{\sqrt{10}}, 2 + \frac{1}{\sqrt{10}} \right) \text{ and}$$

$$\left(-1 + \frac{2}{\sqrt{10}}, 2 - \frac{1}{\sqrt{10}} \right)$$

Eccentricity:

Eccentricity of 1 is

$$e = \frac{c}{a} \Rightarrow e = \frac{\sqrt{5/6}}{1/\sqrt{2}} \Rightarrow e = \sqrt{\frac{5}{3}}$$

Q.2 Show that

(i) $10xy + 8x - 15y - 12 = 0$ and

(ii) $6x^2 + xy - y^2 - 21x - 8y + 9 = 0$

each represents a pair of straight lines and find an equation of each line.

(i)

Solution:

$$10xy + 8x - 15y - 12 = 0 \quad (1)$$

Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 0, b = 0, 2h = 10, 2g = 8 \Rightarrow g = 4,$$

$$2f = -15 \Rightarrow f = \frac{-15}{2}, c = -12$$

(1) represents a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Consider

$$\Delta = \begin{vmatrix} 0 & 5 & 4 \\ 5 & 0 & \frac{-15}{2} \\ 4 & \frac{-15}{2} & -12 \end{vmatrix}$$

$$\Delta = 0 \begin{vmatrix} 0 & \frac{-15}{2} \\ \frac{-15}{2} & -12 \end{vmatrix} - 5 \begin{vmatrix} 5 & \frac{-15}{2} \\ 4 & -12 \end{vmatrix} + 4 \begin{vmatrix} 5 & 0 \\ 4 & \frac{-15}{2} \end{vmatrix}$$

$$\Delta = 0 - 5(-60 + 30) + 4 \left(\frac{-75}{2} - 0 \right)$$

$$\Delta = 150 - 150 = 0$$

Hence (1) represents pair of straight lines

Now we find the pair of lines.

(1) can be written as

$$(10xy - 15y) + (8x - 12) = 0$$

$$5y(2x - 3) + 4(2x - 3) = 0$$

$$(5y + 4)(2x - 3) = 0$$

$2x - 3 = 0$ and $5y + 4 = 0$ are the

required equations of straight lines

(ii) $6x^2 + xy - y^2 - 21x - 8y + 9 = 0 \quad (1)$

Solution: Comparing (1) with

$$ax^2 + by^2 + 2gx + 2fy + 2hxy + c = 0,$$

we get

$$a = 6, b = -1, 2h = 1 \Rightarrow h = \frac{1}{2}$$

$$2g = -21 \Rightarrow g = -\frac{21}{2},$$

$$2f = -8 \Rightarrow f = -4, c = 9$$

(1) represents a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Consider

$$\Delta = \begin{vmatrix} 6 & \frac{1}{2} & -\frac{21}{2} \\ \frac{1}{2} & -1 & -4 \\ -\frac{21}{2} & -4 & 9 \end{vmatrix}$$

Expanding from R_1 , we get

$$\Delta = 6 \begin{vmatrix} -1 & -4 \\ -4 & 9 \end{vmatrix} - \frac{1}{2} \begin{vmatrix} \frac{1}{2} & -4 \\ -\frac{21}{2} & 9 \end{vmatrix} - \frac{21}{2} \begin{vmatrix} \frac{1}{2} & -1 \\ -\frac{21}{2} & -4 \end{vmatrix}$$

$$\Delta = 6(-9-16) - \frac{1}{2}\left(\frac{9}{2}-42\right) - \frac{21}{2}\left(-2-\frac{21}{2}\right)$$

$$\Delta = 6(-25) - \frac{1}{2}\left(\frac{-75}{2}\right) - \frac{21}{2}\left(\frac{-25}{2}\right)$$

$$\Delta = -150 + \frac{75}{4} + \frac{525}{4}$$

$$\Delta = -150 + 150 = 0$$

Hence (1) represents a pair of straight lines

Next we find pair of straight lines

(1) can be written as

$$y^2 - xy + 8y - 6x^2 + 21x - 9 = 0$$

$$y^2 - (x-8)y - 3(2x^2 - 7x + 3) = 0$$

It is quadratic in y, so

$$y = \frac{(x-8) \pm \sqrt{(x-8)^2 - 4(1)(-3)(2x^2 - 7x + 3)}}{2}$$

$$y = \frac{(x-8) \pm \sqrt{x^2 - 16x + 64 + 12(2x^2 - 7x + 3)}}{2}$$

$$y = \frac{(x-8) \pm \sqrt{x^2 - 16x + 64 + 24x^2 - 84x + 36}}{2}$$

$$y = \frac{(x-8) \pm \sqrt{25x^2 - 100x + 100}}{2}$$

$$y = \frac{(x-8) \pm 5\sqrt{x^2 - 4x + 4}}{2}$$

$$y = \frac{(x-8) \pm 5\sqrt{(x-2)^2}}{2}$$

$$y = \frac{(x-8) \pm 5(x-2)}{2}$$

$$y = \frac{x-8+5(x-2)}{2}, y = \frac{x-8-5(x-2)}{2}$$

$$y = \frac{6x-18}{2}, y = \frac{-4x+2}{2}$$

$$y = 3x-9, y = -2x+1$$

Hence pair of straight lines are

$$3x - y - 9 = 0 \text{ and } 2x + y - 1 = 0$$

Q.3 Find an equation of the tangent to each of the conics at the indicated point.

(i) $3x^2 - 7y^2 + 2x - y - 48 = 0$ at (4,1)

Solution: Equation of the tangent at point (x_1, y_1) is given by

$$3xx_1 - 7yy_1 + 2\left(\frac{x+x_1}{2}\right) - \left(\frac{y+y_1}{2}\right) - 48 = 0$$

Put $x_1 = 4, y_1 = 1$

$$3x(4) - 7y(1) + (x+4) - \left(\frac{y+1}{2}\right) - 48 = 0$$

$$12x - 7y + x + 4 - \frac{1}{2}(y+1) - 48 = 0$$

$$13x - 7y + 4 - \frac{1}{2}(y+1) - 48 = 0$$

$$26x - 14y + 8 - y - 1 - 96 = 0$$

$26x - 15y - 89 = 0$ is the required equation of tangent

(ii) $x^2 + 5xy - 4y^2 + 4 = 0$ at $y = -1$

Solution:

First we find x for given $y = -1$

Put $y = -1$ in given equation, we get

$$x^2 - 5x - 4 + 4 = 0$$

$$x^2 - 5x = 0 \Rightarrow x(x-5) = 0$$

$$x = 0, x = 5$$

Hence points of contact are $A(0, -1)$ and $B(5, -1)$

Now equation of tangent to given conic at point (x_1, y_1) is

$$xx_1 + 5\left(\frac{xy_1 + yx_1}{2}\right) - 4yy_1 + 4 = 0 \text{ ..(1)}$$

For $A(0, -1)$, put $x_1 = 0, y_1 = -1$

in (1)

$$x(0) + 5\left(\frac{x(-1) + y(0)}{2}\right) - 4y(-1) + 4 = 0$$

$$\frac{-5x}{2} + 4y + 4 = 0$$

$$-5x + 8y + 8 = 0$$

$5x - 8y - 8 = 0$ is equation of tangent at $A(0, -1)$

For $B(5, -1)$, put $x_1 = 5, y_1 = -1$

in (1)

$$x(5) + 5\left(\frac{x(-1) + y(5)}{2}\right) - 4y(-1) + 4 = 0$$

$$10x - 5x + 25y + 8y + 8 = 0$$

(multiplying by 2)

$5x + 33y + 8 = 0$ is the required equation of tangent at $B(5, -1)$

(iii) $x^2 + 4xy - 3y^2 - 5x - 9y + 6 = 0$ at $x = 3$

Solution:

First we find y for given $x = 3$

Put $x = 3$ in given equation, we get

$$(3)^2 + 4(3)y - 3y^2 - 5(3) - 9y + 6 = 0$$

$$9 + 12y - 3y^2 - 15 - 9y + 6 = 0$$

$$-3y^2 + 3y = 0$$

$$-3y(y - 1) = 0$$

$$y = 0, y = 1$$

Hence points of contact are $A(3, 0)$ and $B(3, 1)$

Now equation of tangent to given equation at point (x_1, y_1) is

$$xx_1 + 4\left(\frac{xy_1 + yx_1}{2}\right) - 3yy_1 - 5\left(\frac{x + x_1}{2}\right)$$

$$- 9\left(\frac{y + y_1}{2}\right) + 6 = 0 \quad (1)$$

For $A(3, 0)$, put $x_1 = 3, y_1 = 0$ in (1)

$$x(3) + \frac{4}{2}(x(0) + y(3)) - 3y(0) - \frac{5}{2}(x + 3)$$

$$- \frac{9}{2}(y + 0) + 6 = 0$$

$$6x + 12y - 5x - 15 - 9y + 12 = 0$$

(multiplying by 2)

$x + 3y - 3 = 0$ is the equation of tangent at $A(3, 0)$

For $B(3, 1)$, put $x_1 = 3, y_1 = 1$ in (1)

$$x(3) + \frac{4}{2}(x(1) + y(3)) - 3y(1) - \frac{5}{2}(x + 3)$$

$$- \frac{9}{2}(y + 1) + 6 = 0$$

$$6x + 4x + 12y - 6y - 5x - 15 - 9y + 12 = 0$$

(multiplying by 2)

$5x - 3y - 12 = 0$ is the required equation of tangent at $B(3, 1)$